LESSON PLAN

L No	TOPICS	Course Outcome Addressed
1	Introduction to the Course.	
2	Periodic functions, Fourier Series, Dirichlet's conditions, Euler's formula for Fourier coefficients.	CO1
3	Even and odd functions. Fourier coefficients of even and odd functions.	CO1
4	Fourier series of functions with period 2π and arbitrary period $2l$.	CO1
5	Tutorial.	
6	Half range sine and cosine series.	CO1
7	Harmonic analysis.	CO1
8	Fourier transforms, basic properties and Parseval's identity and applications.	CO1
9	Tutorial.	CO1
10	**SDL: Fourier sine and cosine transform. Its application in solving heat and wave equations.**	CO1
11	Eigenvalues and eigenspace. Linearly independent eigenvectors.	CO2
12	Diagonalization of a matrix.	CO2
13	Tutorial.	CO2
14	Singular values and singular vectors. Singular value decomposition.	CO2
15	**SDL: Cholesky decomposition**	CO2
16	Tutorial.	CO2
17	System of linear equations, LU decomposition.	CO3
18	QR decomposition.	CO3
19	Recapitulation of basic concepts of vector space including basis and related theorems.	CO3
20	**SDL: Independent/dependent vectors in the space of polynomials over real numbers and in the space of matrices over real numbers. Linear combination of vectors and linear span.**	CO3
21	Dimension of a vector space, examples. Expression for the dimension of the sum of two finite dimensional subspace.	CO3
22	Tutorial.	CO3
23	Linear transformations, kernel and image of a linear transformation.	CO4
24	Rank-Nullity theorem.	CO4
25	Matrix representation of a linear transformation.	CO4
26	Change of basis theorem and related problems.	CO4
27	Tutorial.	CO4
28	Inner product space. Examples of inner product space.	CO5
29	Norm of a vector and related properties.	CO5
30	**SDL: Distance, angle between two vectors and related problems.**	CO5
31	Cauchy-Schwarz inequality	CO5
32	Inner product of functions and orthogonality.	CO5
33	Tutorial.	CO5
34	Tutorial.	CO5
35	Orthogonal complements and orthogonal projections.	CO5
36	**SDL: Construction of orthonormal basis.**	CO5